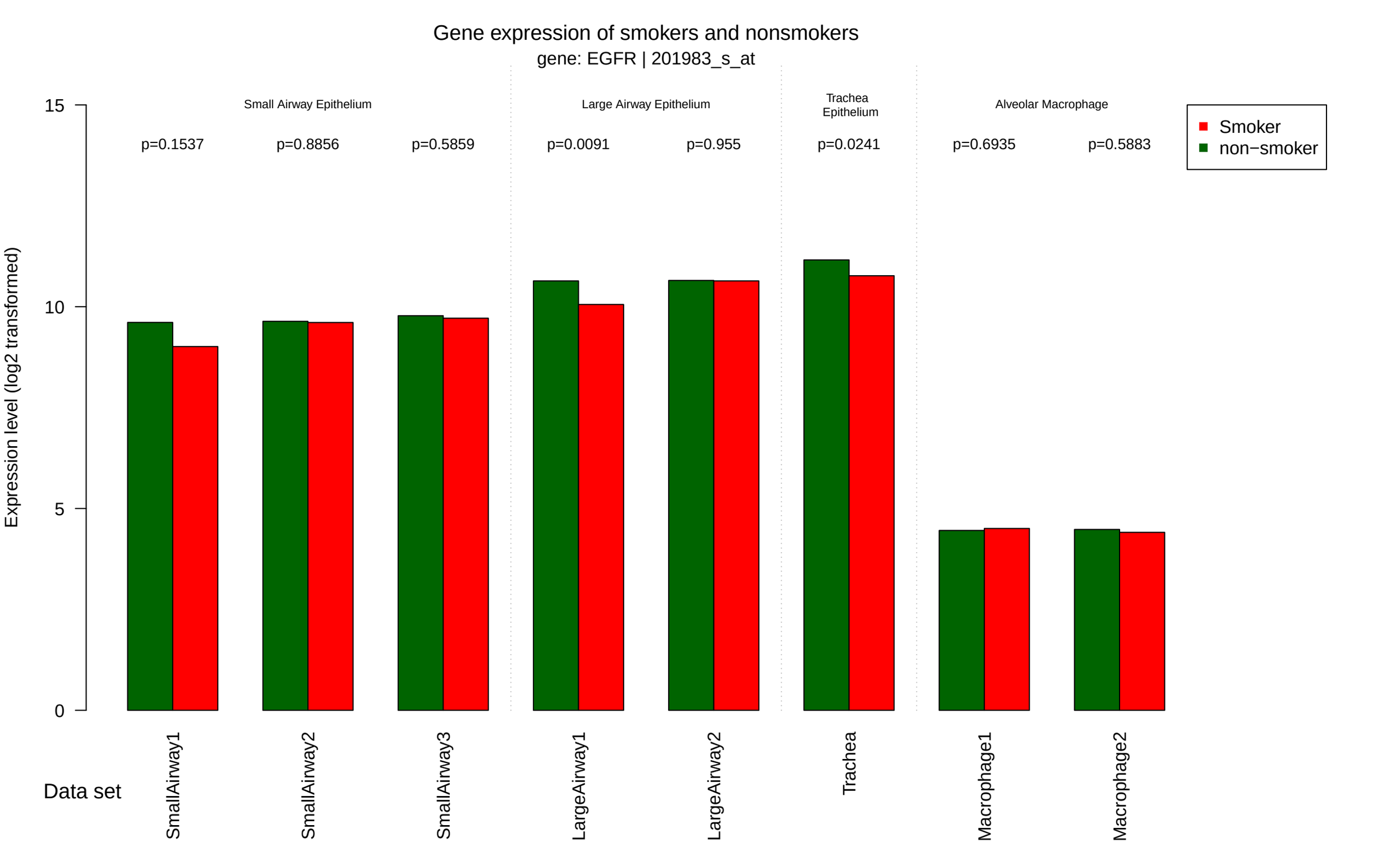
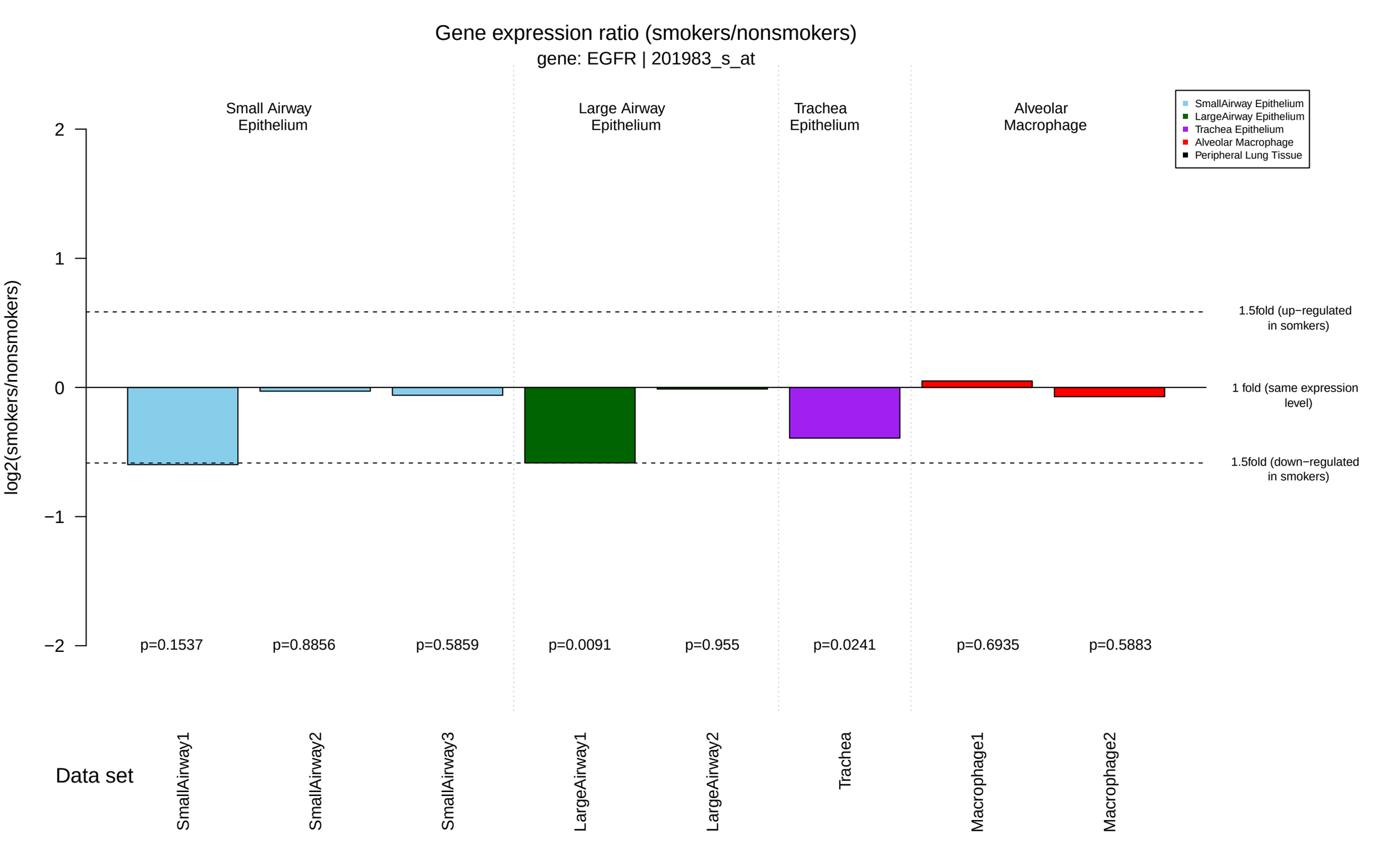
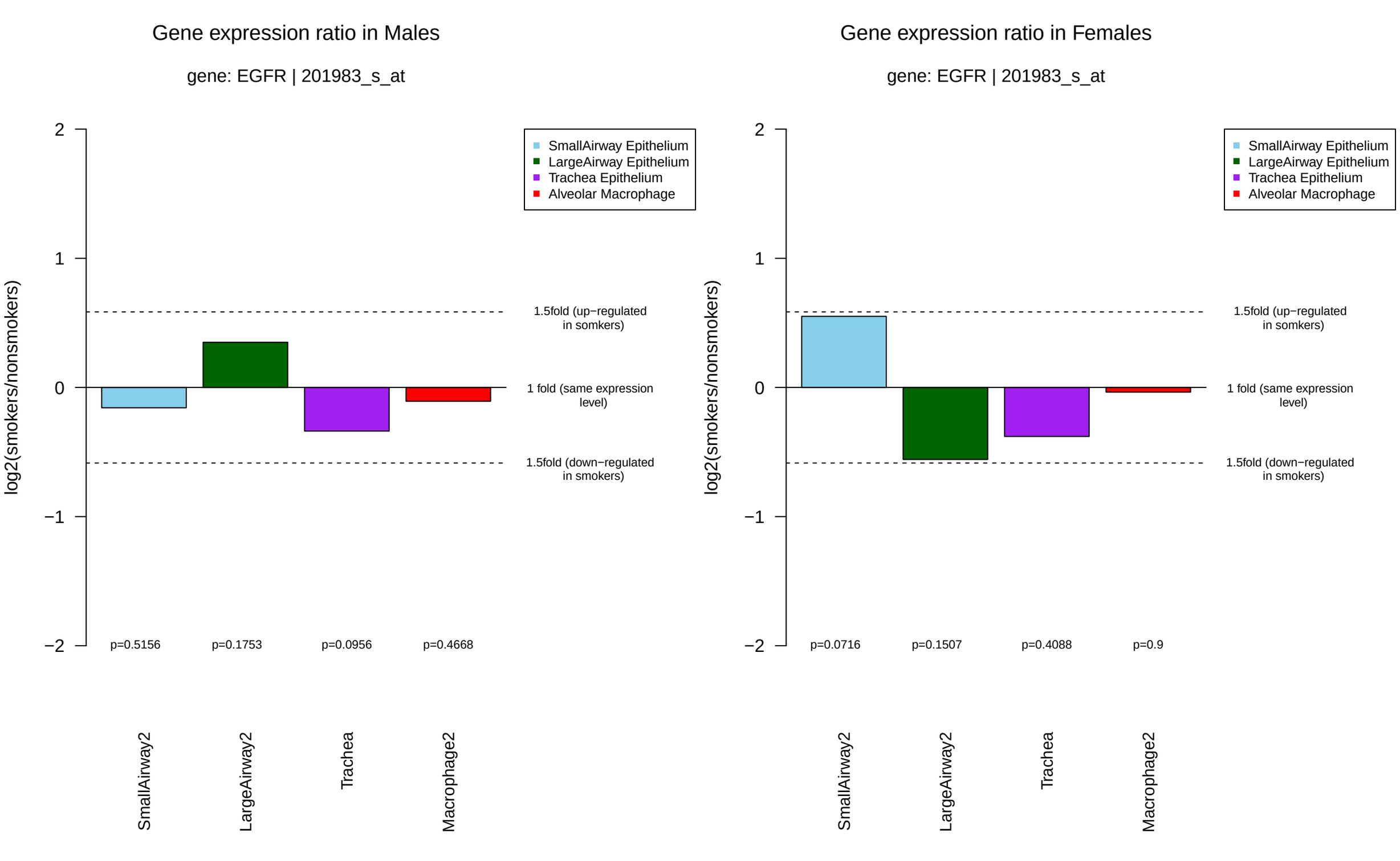
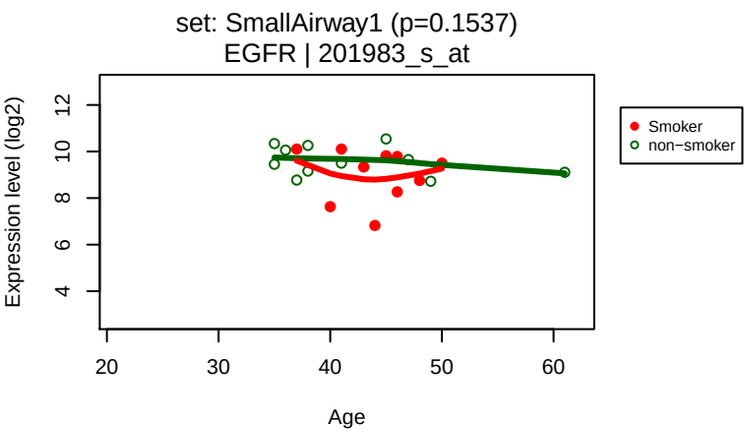
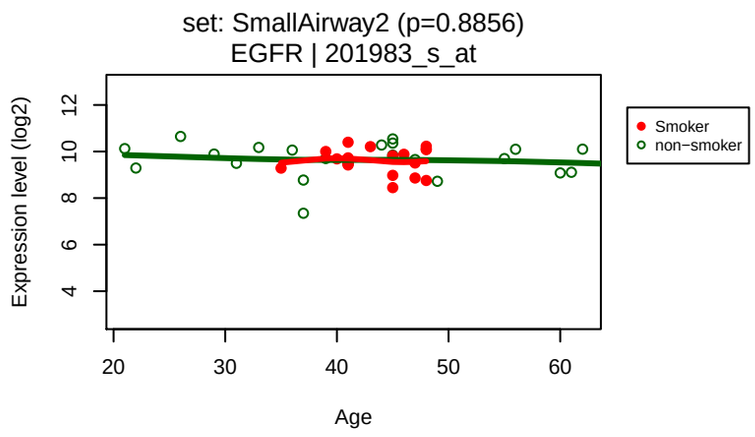
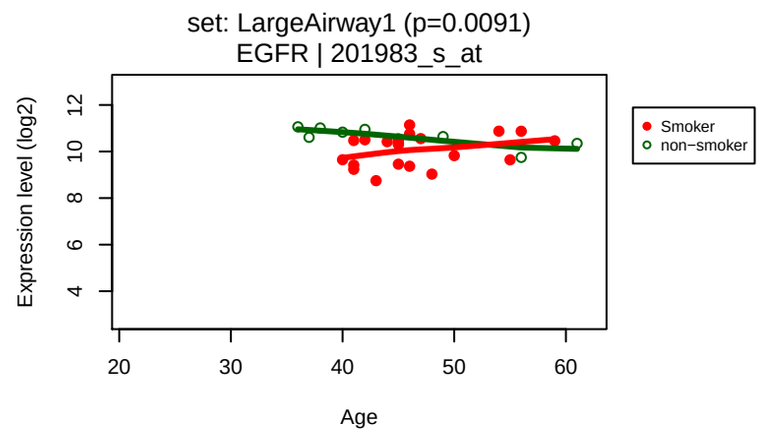

Gene expression of smokers and nonsmokers
gene: EGFR | 201983_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1537
p=0.8856
p=0.5859
p=0.0091
p=0.955
p=0.0241
p=0.6935
p=0.5883
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

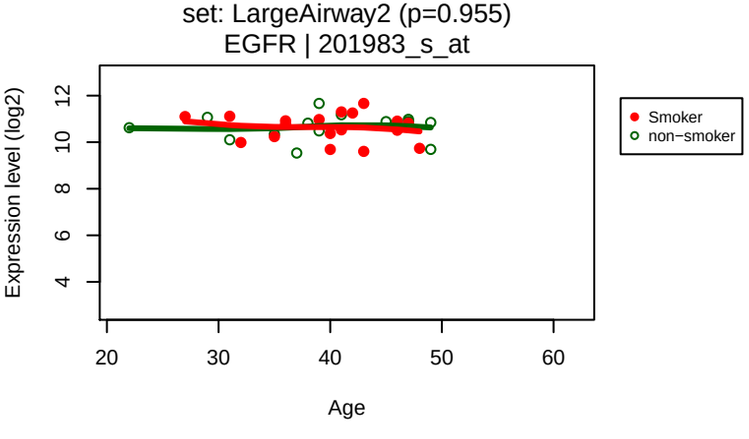

Gene expression ratio (smokers/nonsmokers)
gene: EGFR | 201983_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1537
p=0.8856
p=0.5859
p=0.0091
p=0.955
p=0.0241
p=0.6935
p=0.5883
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

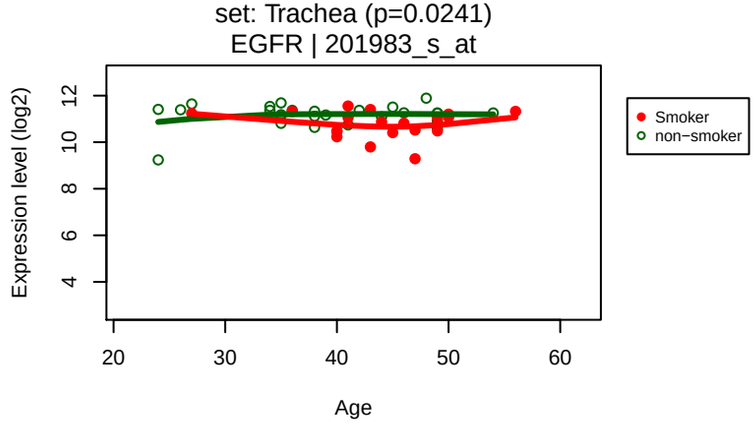

Gene expression ratio in Males
gene: EGFR | 201983_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5156
p=0.1753
p=0.0956
p=0.4668
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EGFR | 201983_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0716
p=0.1507
p=0.4088
p=0.9
SmallAirway2
LargeAirway2
Trachea
Macrophage2

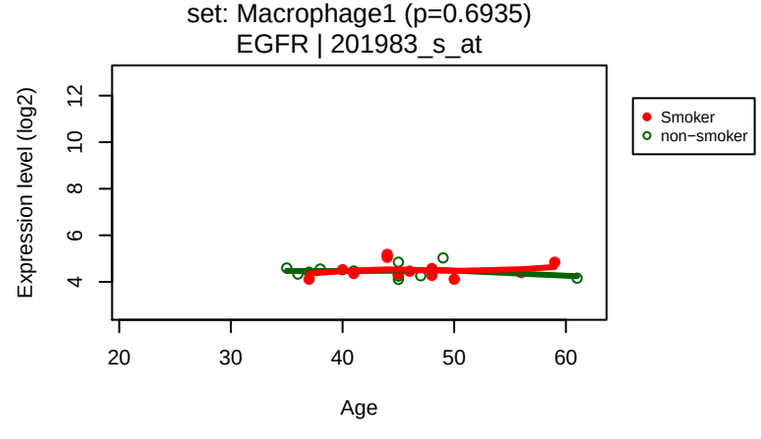

set: SmallAirway1 (p=0.1537)
EGFR | 201983_s_at
Expression level (log2)
Age
Smoker
non−smoker

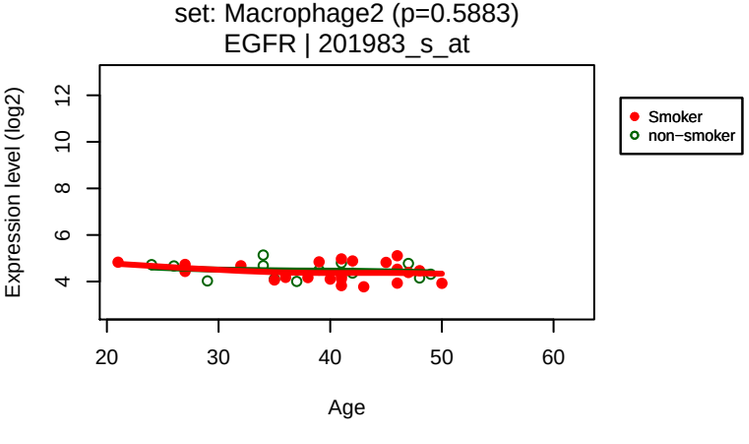

set: SmallAirway2 (p=0.8856)
EGFR | 201983_s_at
Expression level (log2)
Age
Smoker
non−smoker

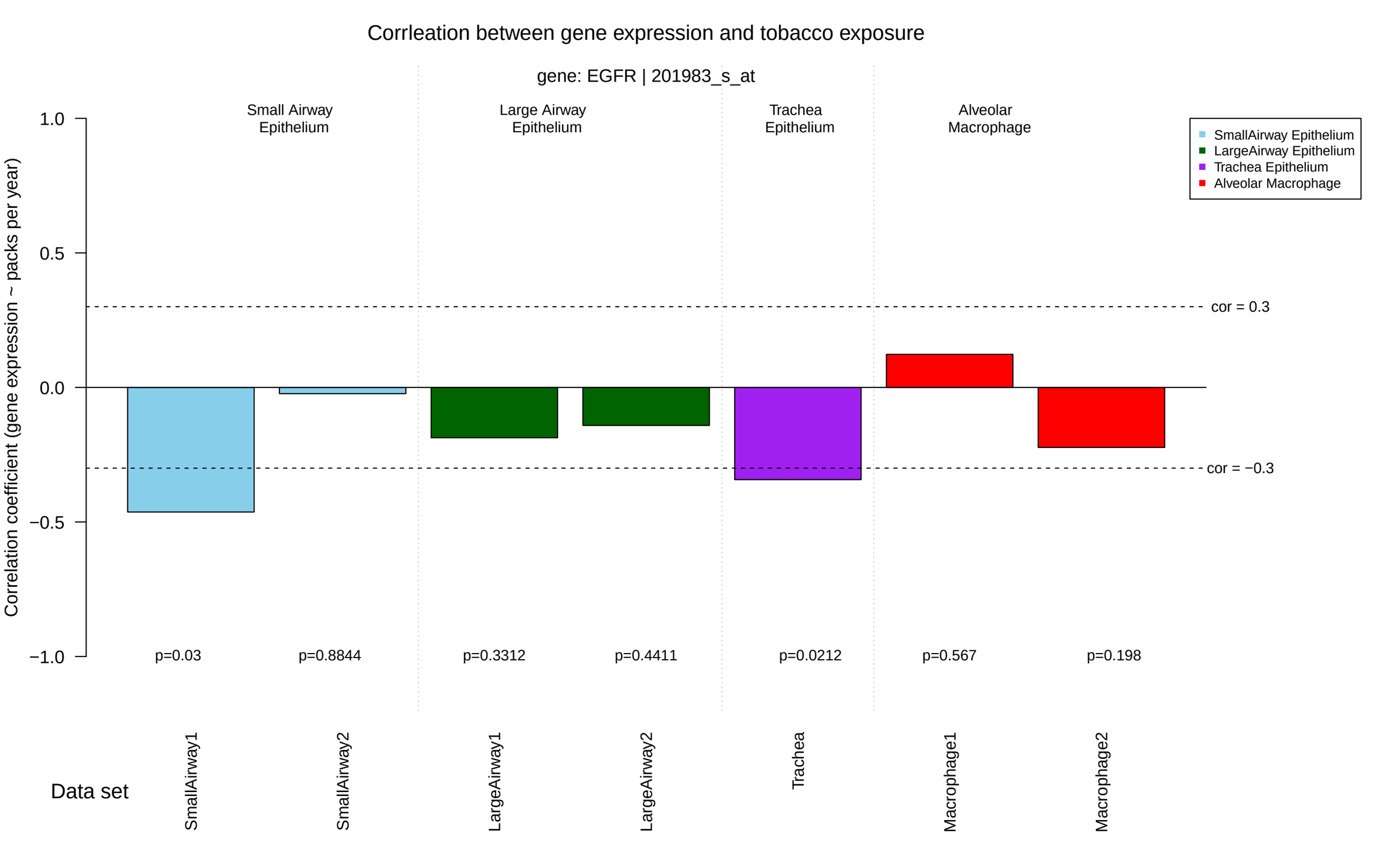

set: LargeAirway1 (p=0.0091)
EGFR | 201983_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.955)
EGFR | 201983_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0241)
EGFR | 201983_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6935)
EGFR | 201983_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5883)
EGFR | 201983_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EGFR | 201983_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.03
p=0.8844
p=0.3312
p=0.4411
p=0.0212
p=0.567
p=0.198
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2